MATHEMATICS – CLASS X

2014

1. Answer all the questions (1X6=6)

(i) Smallest composite number is:

(a) 2 (b) 4 (c) 6 (d) 1.

(ii) Solution of the equation $\frac{x^3}{x}=1$ are:

(a) 0, 1 (b) 0, -1 (c) 1, -1 (d) 0, 1, -1.

(iii) Which one will be the larger between the following?

(a) Ratio of 8 m and 10 m.

(b) 20% 0f $4\frac{2}{5}$.

(iv) Find the value of $\sqrt{8}\times 3\times\sqrt{2}$.

(v) The lengths of the sides of a triangle are 8 cm, 15 cm and 17 cm respectively. Is it a right angles triangle?

(vi) Express 18^0 in radian.

2. Answer all the questions. (2X7=14)

(i) For what rate of simple interest in 5 years will the interest be $\frac{1}{4}th$ of the principal?

(ii) Ratio of the cost price and the selling price is 10 : 11. Find percentage of the profit.

(iii) If x, y are two positive integers such that $x>2,\ y<3,\ 3x+4y\geq 6$, write down the solution set of inequation in the given table:

X		
y		

(iv) Match the following statement in left column with that of the right column (any TWO).

LEFT	RIGHT
(a) Cyclic Parallelogram is a _______.	(i) Proportional
(b) If two circles touch each other internally, the distance between two centres is equal to the ______ of the lengths of the radii of two circles.	(ii) Right angle
(c) Corresponding sides of two similar triangles are _____________.	(iii) Rectangle
(d) Angle in a semicircle is a _______.	(iv) difference

(v) ABC is an equilateral triangle having each side of length 6 cm. G is centroid of the triangle. Find length of AG.

(vi) The volume of a right circular cone is X cubic unit, area of the base is Y sq unit and height is Z unit. Find $\frac{YZ}{X}$.

(vii) If $\cos\alpha = \sin\beta$, where α, β are acute angles, find the value of $\sin(\alpha + \beta)$.

3. Answer any TWO. (5X2=10)

(a) Two similar pots of equal volume had $\frac{1}{2}$ and $\frac{1}{3}$ parts respectively filled with milk. The remaining empty part of these pots was filled with water. The contents of both the pots were poured into another pot. What will be the ratio of milk and water in the new pot?

(b) A farmer deposited some money in the village post office. After 4 years he came to know that the amount of his money became Rs. 434. He calculated and observed that the interest received by him was $\frac{6}{25}$ part of the principal. Find the sum of the money deposited and annual rate of simple interest in the post office.

(c) At a rate of 8% interest per annum, find the compound interest on Rs. 10000 for 9 months, compounded quarterly (every 3 months).

(d) A publisher requires paper worth Rs. 3875, printing cost of Rs. 3315 and binding cost or Rs. 810 to publish 2000 copies of a book. Find the marked price of each book such that after giving 20% commission to the book seller, the publisher gets 20% profit.

4. Answer any ONE question: (4X1=1)

(a) Find HCF of: $3x^3 - 24, \quad x^4 - 4x^3 + 4x^2, \quad x^3 - 5x^2 + 6x$

(b) Find LCM of: $12(x^2 + xy)^2, \quad 4(xy - y^2), \quad 6(x^2y^2 - y^4)$

5. Solve any ONE (3X1=3)

(a) $x - \dfrac{2}{y} = 24, \quad x - \dfrac{8}{y} = 48$, [Using method of elimination or of cross multiplication.]

(b) $\dfrac{x}{x+1} + \dfrac{x+1}{x} = 2\dfrac{1}{12}$.

6. Answer any ONE question: (4X1=4)

(a) If 2 is added to both numerator and denominator of a fraction, it becomes $\dfrac{9}{11}$ and again the fraction becomes $\dfrac{3}{4}$ when 1 is subtracted from both numerator and denominator. Find the fraction.

(b) If the positive square root of a positive integer is subtracted from the same positive integer gives 110, find the positive integer.

7. Draw the graphs of the inequalities and indicate the solution region (any ONE). (4X1=4)

(a) $x \geq 4, \quad y \geq 1, \quad 3x + 5y \leq 30$.

(b) $x \geq 0, \quad y \geq 0, \quad x + y \geq 10$.

8. Answer any ONE question: (3X1=3)

(a) If $x = cy + bz,\ y = az + cx,\ z = bx + ay$, prove that $\dfrac{x^2}{1-a^2} = \dfrac{y^2}{1-b^2}$.

(b) If $\dfrac{a^2}{b+c} = \dfrac{b^2}{c+a} = \dfrac{c^2}{a+b} = 1$, then show that $\dfrac{1}{1+a} + \dfrac{1}{1+b} + \dfrac{1}{1+c} = 1$.

9. Answer any ONE question: (3X1=3)

(a) If $\frac{x}{y} \propto (x+y)$ and $\frac{y}{x} \propto (x-y)$ then show that $x^2 - y^2$ is constant.

(b) The volume of a sphere varies directly as cube of its radius. Three solid spheres of radii 3 cm, 4 cm and 5 cm are melted and a new solid sphere is made. If no volume is lost for melting, find the diameter of the new sphere.

10. Answer any ONE question: (3X1=3)

(a) If $x = \frac{1}{2+\sqrt{3}}$; $y = \frac{1}{2-\sqrt{3}}$, then find the value of $\frac{1}{1+x} + \frac{1}{1+y}$.

(b) If $2x = \sqrt{5}+1$, show that $x^2 - x - 1 = 0$.

11. Answer any TWO questions: (5X2=10)

(a) Prove that, if a straight line divides the two sides of a triangle proportionally, then the straight line is parallel to the third side of the triangle.

(b) Prove that, the segments of two tangents of a circle from an external point to the points of contact are equal and they subtend equal angles at the centre.

(c) Prove that, if the area of the square on one side of a triangle be equal to the sum of areas of the squares on the other two sides, the angle opposite to the greatest side being right angle.

12. Answer any ONE question: (3X1=3)

(a) The centre of a circle has been removed. Then using what geometrical method, you will restore that? (Answer by means of construction.)

(b) ABC is a right angled triangle whose $\angle A$ is 90^0. AD is perpendicular on the hypotenuse BC. Prove that $\frac{\Delta ABC}{\Delta ACD} = \frac{BC^2}{AC^2}$.

13. Answer any ONE question: (5X1=5)

(a) Find geometrically the value of $\sqrt{21}$. (only traces of construction are required)

(b) Triangle ABC is such that AB = 8 cm, BC = 6 cm and $\angle ABC = 60^0$; then draw the triangle and also draw the circumcircle of the triangle.

14. Answer any ONE question: (3X1=3)

(a) Three small spheres are made by melting a solid lead sphere of diameter 12 cm. The diameters f the small spheres are in the ratio 3 : 4 : 5. Find the radius of each small speres.

(b) The ratio of the height and radius of the base of a right circular cylinder is 3 : 1 and volume of that cylinder is 1029π cubic cm. Find the total surface area of the cylinder.

15. Answer any ONE question: (4X1=4)

(a) Two solid spheres of radii 1 cm and 6 cm are melted and a hollow sphere of thickness 1 cm is made. Find the outer curved surface area of the new sphere.

(b) A cylinder with diameter 12 cm is partly filled with water. A solid spherical metal ball is dropped into the cylinder. Thereafter the level of the water rises by 1 cm. Find the diameter of the ball.

16. Answer any TWO questions: (3X2=6)

(a) Find the value of $\cot^2 30^0 - 2\cos^2 60^0 - 4\sin^2 30^0 - \frac{3}{4}\sec^2 45^0 + \tan 45^0$.

(b) If $\angle A + \angle B = 90^0$, then show that $1 + \frac{\tan A}{\tan B} = \tan^2 A \sec^2 B$.

(c) Solve: $\frac{\sin\theta - \cos\theta}{\sin\theta + \cos\theta} = \frac{1-\sqrt{3}}{1+\sqrt{3}}$.

(d) In a right angled triangle the difference between two acute angles is $\frac{2\pi}{5}$. Express these angles in terms of radian and degree.

17. Answer any ONE question: (5X1=5)

(a) From a point A on the ground the angle of elevation of the top of the pillar is 30^0. If the point A moves 20 m towards the foot of the pillar and reaches at the point B, then angle of elevation becomes 60^0. Find the height of the pillar and the distance from A to the pillar.

(b) The heights of two towers are 180 m and 160 m respectively. If the angle of elevation of the top of the second tower from the foot of the first tower is 30^0, what is the angle of elevation of the top of the first tower from the foot of the second.

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